

RESPONSE TO THE REFEREE ON NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

RESPONSE TO THE REFEREE

We thank the referee for focusing the revision on the economic contribution of the learned continuation. The title and subject remain *Neural Bellman Operators*. The revised organization makes the economic value of reusable continuation the central question, retains the original applications in a current companion, and preserves every mathematical statement and proof in the current scientific documents.

The revision completes four separately frozen scientific families: the five-method continuation menu, independent confirmation of all original policies, newly trained economic robustness with a strengthened HJB procedure, and a signed Bellman assessment. Each report retains its full declared population and confidence family. The main results now state the actual economic comparisons and measured work; the technical supplement preserves complete current proofs, diagnostics, and all reported design cells.

The review considered publication commit 1cb3cc9e and is pinned at review commit cb4595bb. It is a repository-owner-commissioned advisory report, not a journal decision. We distinguish the previous post-freeze analytical reanalysis from independent prospective payoff confirmation and distinguish both from a new economic decision experiment.

1. BLOCKING COMMENTS

B1. The learned Bellman critic still has no demonstrated incremental economic value

The revised paper asks directly when reuse of a learned continuation improves economic decisions or complete computational work. Section 4.4 states the shared Bellman object and proves its cross-task identity in Proposition S.14. Section 5.1 compares five complete reusable procedures, allowing the Raw alternatives to cache the information permitted by the same queries. The work boundary includes cache

construction, every declared fitting and action-solver stage, query evaluation, and independent economic assessment. The title remains *Neural Bellman Operators*; the direct continuation comparison supplies the central empirical question.

The earlier NBO–Raw and NBO–DPO intervals remain unresolved at the stated margin, and Raw’s lower complete work at the fifty-dimensional online target remains visible in Section 6.5. A gain over the schedule, a sharper deterministic transfer, or superiority to one HJB implementation is not presented as a substitute for incremental value of the learned continuation.

The completed menu provides the direct result. At the final prescribed stage, NBO’s lower payoff advantage over the materialized Raw actor is 0.001802 in Long and 0.000147 in Intermediate, both in dimension fifty and rounded downward. The corresponding lower advantages over direct policy are 0.000327 and 0.000130. All four exceed the prospectively fixed margin 10^{-4} . Neither of these economic cells had an outcome pilot. NBO also uses less complete construction and query work than direct policy in both cells. Table I reports every final-cell direct comparison, and the technical supplement retains all three budgets and all 216 protected endpoints.

The Raw alternatives remain distinct. Cached SAA attains economically equivalent final-budget payoffs within $\pm 10^{-4}$ in six of eight cells, where NBO uses less construction and query work. SAA has materially higher payoffs in both Long cells. The vector and Raw actors are cheaper than NBO in every final cell. The contribution is therefore a demonstrated economic benefit of the declared fitted-continuation procedure in specific repeated decisions, together with explicit cost and accuracy comparisons. It is not an inferred reduction in independently measured critic risk or an assertion that NBO is the cheapest possible Raw implementation.

B2. The neural-HJB result is specific to one difficult and insufficiently validated implementation

Section 5.3 specifies a stronger comparator with an exact diffusion trace at dimensions ten and fifty. Four predetermined learning-rate and collocation configurations use Adam and L-BFGS. Independent residual banks select among their retained checkpoints; final policy payoffs do not select the network. The implementation

retains both an optimized greedy-action root and a distilled actor and charges all fitting, selection, distillation, and deployment work. Residual-based early stopping within a fixed envelope is distinguished from the earlier online economic-target rule.

The previous stochastic-trace and 44-step action-search implementation remains in technical Section S.14, with its original outcomes and costs. Any new ranking concerns the declared strengthened procedures. Exact network derivatives and independent residual selection strengthen this implementation comparison; they are not an independent value-error theorem for HJB or a ranking of the entire method class.

The full 64-trial comparison is complete. All forty pooled method gains exceed the primary schedule-relative target .0005. Table VI reports every direct interval: only the low-volatility, fifty-dimensional NBO comparisons against HJB’s greedy and distilled deployments clear 10^{-4} , with downward-rounded lower endpoints .000334 and .000340. The other thirty direct comparisons cross zero and remain unresolved, including both strengthened-HJB comparisons in the original high-volatility calibration. We consequently keep these outcomes separate from replication against the original HJB implementation.

Technical Section S.21 reports all four configurations, independent residual and action diagnostics, trace checks, and complete work. A terminal lift enforces the terminal condition; it is not independent evidence of global value accuracy. Successive value and policy changes compare iterates. No direct greedy-minus-distilled payoff event was registered, so their separate outcomes are not described as a protected distillation-loss comparison.

B3. The dimension-10 superiority claim is a post-freeze analytical reanalysis

We label the R15 dimension-ten result a post-freeze analytical reanalysis. The original NBO-minus-HJB interval included zero. The paired-transfer refinement was developed after the original analytical freeze and inspection of interval widths, on the same simulated observations. Both original and refined endpoints remain current in Section 6.5 and the complete paired table in the technical supplement.

The refinement did not change the original policies, confidence event, or online stopping decisions.

Section 5.2 defines a separate confirmation on all 128 frozen policies. Its paired proof, protected constants, reporting decisions, and family allocation of 0.01 are fixed before the new bank of 8,192 paths on 2,048 cells. The independent derivation and implementation audit is in technical Section S.7.6; it retains the original outward bounds even where a high-precision diagnostic is smaller.

The new bank is complete. Table V gives downward-rounded NBO-minus-original-HJB lower endpoints of 0.000176 and 0.001342 in dimensions ten and fifty. Both exceed the fixed materiality margin 10^{-4} . The original-transfer dimension-ten interval on this new bank still includes zero. The four NBO–Raw and NBO–direct-policy comparisons remain unresolved. These are prospective conditional payoff comparisons over the unchanged sixteen-stream policy population; the earlier R15 analysis retains its post-freeze label.

B4. The finite method distribution and calibration have narrow external scope

The main article states the finite method distribution and the economic population separately. The earlier sixteen streams are fully enumerated members of the declared algorithm distribution. A new payoff bank on those policies provides conditional replication; it does not add training configurations or demonstrate robustness to a different calibration. The stylized balance-sheet profiles and fee experiment remain model experiments rather than an empirical calibration.

The separate robustness family trains all methods anew at four continuous-diffusion calibrations, in both dimensions and eight new streams per cell. Its high, low, longer-horizon, and joint-stress designs differ from the four finite menu calibrations. The stress design changes multiple primitives, initial heterogeneity, and the feasible actor radius together. Its result is interpreted as performance at that joint economic design, without single-primitive causal attribution.

The repeated-decision study specifies four calibrations, both dimensions, twelve tasks, and 384 fixed task-state queries in Section 5.1. The Original, Quarterly, and Intermediate calibrations have quarter-year periods; Long has half-year periods.

Quarterly and Long each have eight periods. Quarterly had one disclosed exploratory pilot; Long, Intermediate, and the three tighter-cap tasks had no outcome pilot. Technical Section S.19 retains this development history. These finite objectives are distinguished from the continuous-diffusion payoff and full-adapted-class target of Section 4. The original recursive-utility, preference, temporal-self, and game applications remain current in the main scope section and complete application companion.

The completed finite menu retains 128 trials, 640 method fits, and 1,920 method-stage candidates without training failure or fallback. The positive Raw-actor and direct-policy margins occur in the previously unpiloted fifty-dimensional Long and Intermediate cells. Six final-budget SAA comparisons are equivalent at 10^{-4} , and SAA is materially better in the two Long cells. The continuous-economy robustness study independently retains all 64 trials and 320 outputs, with positive pooled schedule gains for all forty method–economy combinations and only two resolved material direct NBO–HJB differences. Tables S.29 and S.39 give the separate economic grids. These are results over the stated grids and complete finite stream populations, not an empirical calibration or a claim about all possible economic parameters.

B5. The high-dimensional result is conservative safe improvement, not an accurate HJB solution

Section 6.5 presents the existing schedule-relative gains beside the full-adapted-class regret upper bounds, 0.025971 and 0.036843 after upward rounding. The latter retain the anchor gap; they are not presented as 10^{-4} high-dimensional optimality certificates. The capital payoff theorem and its complete anchor and diffusion accounts remain in the current main article and technical supplement.

The scalar Howard account retains its distinct finite-grid residual, action-restriction, mesh, and domain statements. Its favorable finite-grid accuracy is not extrapolated to dimension fifty. Likewise, the new finite repeated-decision problem has its own value target. The article retains the original nonlinear control economy and applies each accuracy term to the economic object it actually bounds.

B6. The work comparison does not identify an NBO efficiency advantage

The work comparison charges the reusable continuation and a credible Raw cache symmetrically. Section 5.1 includes construction, fitting, tuning, selection, new queries, action implementation, and verification. Section 5.3 charges all four candidate HJB fits and both deployment implementations. Shared construction costs and any amortization population are reported explicitly.

The earlier complete timings and stopping records remain in the technical supplement. At dimension fifty Raw reached the same original online target with less mean complete work than NBO. The earlier NBO–DPO comparison is retained with its stated complete process boundary. Neither the earlier fixed-work prefix frontier nor a fixed-envelope residual comparison is recast as a new prospective economic work-to-target experiment.

For the new menu, final-stage time is the actual complete three-stage process. Earlier-stage records allocate the executed prefix and all measured outer overhead; they do not observe a separate stopping experiment. Confirmation is a shared five-method, three-stage workload with a separate full bill. Timing summaries describe these runs and do not add a runtime-distribution confidence claim. Every stage remains reported alongside its economic interval.

The final-stage menu gives two concrete comparisons of economic value and work. In the unpiloted Long and Intermediate fifty-dimensional cells, NBO has material protected payoff gains over direct policy and mean complete construction/query times of 14.59 versus 38.03 seconds and 12.70 versus 45.58 seconds, respectively. In the six final-budget cells with NBO–SAA equivalence at 10^{-4} , NBO uses 19.73–84.02% less construction and query work. Table S.34 separately reports the full common confirmation bill for each cell. Charging that same entire bill to both methods is an explicit conservative allocation, not an independently observed single-method verification time.

These comparisons are tied to the prescribed final stage. The four- and sixteen-path SAA budgets remain in the current tables; earlier SAA budgets can already supply cheaper equivalent decisions in some cells. The vector and Raw actors are also cheaper than NBO in every final cell. Accordingly, the paper reports the demon-

strated final-budget economic and work tradeoffs without converting them into a minimum-work-to-margin frontier. Table S.44 likewise retains the strengthened HJB procedure’s full fixed-envelope costs and every final target outcome.

B7. The operational mechanism account is scientifically valuable but does not support the paper’s causal narrative

The new signed account evaluates the economic Bellman identity before constructing separate error endpoints. Proposition S.12 uses a shared future innovation array at the candidate and reference postdecision states. The current reward plus paired continuation-return difference equals predicted improvement minus signed continuation error sample by sample. Proposition S.13 bounds its range using economic primitives and the action tube, without a fitted-network norm. The full derivations remain in technical Section S.7.7.

The original nonpositive mechanism lower bounds and unresolved critic–Raw risk comparisons remain in the current evidence. The new positive signed-policy bounds verify the selected policies’ occupation Bellman account. Their interpretation is separate from attribution of the package’s improvement to the learned critic. The direct continuation comparison in Section 6.1 addresses that separate question.

The complete fresh family comprises 32 original NBO policies and 32,768 continuation bridges. Table VII reports continuous-economy gain lower bounds of 0.000767 and 0.000776, rounded downward. Both exceed the primary 0.0005 target; neither reaches the secondary 0.001 target. The joint signed account therefore supplies positive operational mechanism bounds where the original Cauchy–Schwarz account was nonpositive. All six protected intervals remain in technical Section S.20.1. The signed correction intervals still contain zero, and the earlier critic–Raw risk uncertainty is retained.

B8. The manuscript remains too broad relative to the result

The main argument is now organized around the incremental economic value of reusing a learned continuation. The capital model and operational Bellman account remain in the main article. General mathematical results, complete proofs,

and detailed computational evidence are organized by subject in the technical supplement. Section 7 retains the original applications' economic content in the main article, and the current *Economic Applications* companion provides their complete models, derivations, proofs, and numerical records.

This reorganization retains every one of the reviewed R15 article and supplement's 285 labels in the three current scientific documents. It also checks all 33 explicit mathematical and proof environments and 18 complete proof sections identified by their headings. The original application text is retained with only enumerated location and grammar corrections, and no mathematical changes. Section S.22 identifies the exact source archive and the machine-readable preservation map. The previous negative and unresolved outcomes remain available in the current reading set.

2. MAJOR COMMENTS

M1. Choose a single publication claim

The publication question is the economic value of a reusable Neural Bellman Operator continuation. The introduction, computational design, and direct result section follow that question. The paired-transfer and signed-assessment results support the validity and precision of economic comparisons; they do not replace a critic-specific comparison. The original title and economic applications are retained.

The abstract and conclusion now state the common continuation object, the fixed-dictionary approximation–variance account, the two unpiloted fifty-dimensional material Raw-actor and direct-policy gains, and the six final-budget SAA equivalence comparisons with their actual work scope. They also retain the higher SAA payoffs in both Long cells. The supplementary prospective payoff and signed-account results support this economic question without substituting for its direct simulation comparison.

M2. Strengthen the HJB comparator before making it the positive centerpiece

The strengthened comparator is specified in Section 5.3: exact diffusion trace, four frozen learning-rate and collocation configurations, Adam and L-BFGS, indepen-

dent residual selection, and separately retained greedy and distilled implementa-
tions. Its cost includes every candidate and selection step. The old HJB procedure
remains part of the evidence.

The completed comparison retains all four configurations in every economic
cell, all independent diagnostics, and both deployments. All forty pooled method
gains exceed .0005. Only the low-volatility, fifty-dimensional NBO–HJB greedy and
NBO–HJB actor intervals have lower endpoints above 10^{-4} ; the remaining thirty
direct comparisons are unresolved. The new exact-trace procedure is therefore not
described by the earlier implementation’s ranking. Technical Section S.21 reports
the complete residual, trace, action, and work account. Finite collocation diagnostics
and iterate changes are not global value-error certificates.

M3. Prospectively replicate the paired-transfer result

The paired confirmation fixes its proof, outward constants, candidate identities,
clipping account, reporting code, and 0.01 family before new observations. It uses
all 128 original policies and an independent 8,192-path, 2,048-cell bank. Section 5.2
states this target and technical Section S.7.6 records the separate derivation and
implementation audit.

Section 6.2 reports the completed 32-bank confirmation and all six direct method
comparisons. NBO exceeds the original HJB implementation by the prespecified
 10^{-4} margin in both dimensions; Raw and direct-policy comparisons remain unre-
solved. All 238 events and 128 policy executions are retained, with no replacement
stream or post-outcome sample extension. The target remains the complete origi-
nal finite policy distribution. The new per-policy final certificates and additional
confirmation work are reported separately from original stopping in technical
Section S.20.

M4. Resolve the Raw comparison or change the protagonist

The revised protagonist remains the learned continuation in NBO. The repeated-
decision design compares its reuse directly with retained simulation information

under credible Raw caching. Section 6.5 keeps the earlier unresolved Raw intervals and Raw's lower complete work at the original fifty-dimensional target visible.

Equation (31) and Proposition S.15 also identify exactly when a fixed scalar-gradient dictionary removes more noise variance than it introduces approximation error. This identity is conditional on its fixed features and applies equally to a Raw estimator using that same projection. The nonlinear fit is assessed by its actual economic decisions. The five-method menu compares that complete fitted procedure with the vector predictor, materialized Raw and direct-policy actors, and direct sample-average Raw; no independent critic-risk reduction is inferred from the payoff comparison.

The completed direct Raw-actor comparison is materially positive in the two unpiloted fifty-dimensional cells: the lower endpoints are .001802 and .000147, rounded downward. These positive comparisons concern the fitted representations and action procedures actually executed. Against cached SAA at its prescribed final 64-path budget, six cells attain practical equivalence within $\pm 10^{-4}$ with less construction and query work for NBO; the two Long cells favor SAA materially. All three SAA budgets remain visible, so the paper does not infer a minimum-work advantage from the expensive final budget alone. These findings supply a direct, bounded economic reason to retain NBO as the protagonist, while preserving the earlier Raw results and the cost of fitting the critic.

M5. Expand robustness over economic designs, not only random streams

Section 5 separates variation in economic problems from variation in random streams. Table S.29 gives all four menu calibrations, spanning horizons of one to four years, idiosyncratic volatility from 0.15 to 0.60, common volatility from 0.10 to 0.30, coupling from 0.15 to 0.30, and terminal dispersion penalties from 0.03 to 0.10. Twelve current preference, externality, and cap tasks share the continuation within each calibration. The finite model has its own value target. The separate fresh-bank replication conditions on the original policies and does not supply this economic variation.

The stronger-HJB comparison additionally uses four continuous-economy calibrations and eight newly trained streams per calibration and dimension. This is a second economic design family, with a separate 0.01 allocation for all 648 events. Its new training streams, joint stress variation, and continuous-time transfer are stated separately from the menu's sixteen-stream finite-period population.

All eight finite menu cells and all eight continuous-economy cells are now reported. The menu's material Raw-actor and direct-policy gains include the two unpiloted fifty-dimensional designs, while its final SAA outcomes comprise six equivalent cells and two material reversals. The continuous-economy family gives positive pooled schedule gains in all forty method-economy combinations; only two of its 32 direct NBO contrasts attain the materiality margin. The full tables retain the unsuccessful accuracy certificates and every unresolved comparison. The main conclusions name the relevant designs and distinguish the finite menu's four-year Long calibration from the continuous economy's two-year long calibration.

M6. Tighten or relabel the high-dimensional accuracy claim

The main historical panel places the protected schedule-relative gains beside the full-class regret bounds and states the latter's remaining anchor gap. The original scalar finite-grid calculations retain their residual and refinement interpretation. The new finite-period menu compares the same explicit decision values across all methods; its separate scalar accuracy statement covers a complete continuous intervention interval in that finite economy. Neither scalar target is identified with the high-dimensional adapted optimum. These distinctions appear in Sections 5 and 6.5, while the original full-control problem and mathematical accounts remain current. The new scalar account certifies a gap below 10^{-4} for 80 of 128 fitted policies, covering all sixteen streams in each of five economic cells. Its remaining 48 upper bounds do not certify that tolerance; this is not a lower bound showing that the policies' actual regret exceeds it. The target is the whole continuous common-action interval at the first stored state and central task, not the vector optimum or the continuous-diffusion adapted optimum.

M7. Improve the mechanism experiment's precision

The new assessment uses the direct paired future-return statistic in Proposition S.12. Its range in Proposition S.13 avoids the fitted predictor's full range by retaining the sample-level cancellation. Independent occupation and continuation banks, a power calculation, and a separate account for deterministic transfer and arithmetic determine its precision. The original mechanism and critic-risk intervals remain current.

The completed assessment uses 1,024 bridges for each of 32 policies, with the six-event family fixed at $\alpha = 0.01$. Section 6.4 reports positive continuous-economy gain lower bounds exceeding 0.0005 in both dimensions. Technical Section S.20.1 retains all six M , C , and G intervals and the complete fresh-process work. The range and transfer allowances remain explicit in the main table. The signed correction itself remains unresolved, so the gain conclusion is based on the joint statistic rather than an asserted reduction in predictor risk.

M8. Present the post-freeze refinement with a stronger evidentiary label

The main introduction and Section 6.5 identify the R15 dimension-ten result as post-freeze analytical reanalysis and retain its original zero-covering interval. The completed refinement remains mathematically applicable to its stated observations, but its chronology differs from the new bank's prospective protocol. The new confirmation section states what was frozen before the bank and retains the unchanged policy-distribution scope.

M9. Reduce and reorganize the manuscript

The paper now has four coordinated reading documents: the focused main article, technical supplement, complete economic-applications companion, and this response. Cross-references identify the current location of each result. The original title, models, principal application narratives, complete mathematical statements, and proof arguments are retained. The source inventory and editorial map in Section S.22 check all 285 old labels, every explicit mathematical/proof environment,

and complete heading-based proof sections against the reviewed publication. The exact prior roots are archived with their checked Git blob and SHA256 identities.

M10. Clarify a few evidence semantics

The revised terminology distinguishes a method from a particular implementation, a selected policy from the finite method average, online success from a later final certificate, and schedule-relative gain from full-class regret. In particular, no dimension-ten method reached the original online target; a later pooled HJB lower endpoint above that threshold does not make an individual HJB run successful. The original and refined paired intervals refer to the same observations but different deterministic analyses. New payoff paths on those frozen policies are not new training streams.

The stronger HJB comparison uses residual selection within a fixed envelope, whose work semantics differ from the original online economic-target rule. The new finite repeated-decision problem retains its own objective. The scalar mesh and domain diagnostics retain their finite-grid meaning, and the fee experiment retains its specified resource and log-felicity transformation. Sections 5 and 6.5 state these distinctions in the main article; complete supporting records remain current.